



RACEDAY API

Endpoint Plan — Student Edition

A clean, role-based REST API for managing races, categories, enrolments, results and payments.

Two Roles. One Platform.

ORGANISER	Creates events, manages categories, records results, views all enrolments and audit logs.
PARTICIPANT	Browses events, enrolls in categories, pays entry fees, and tracks personal results.

HTTP method colour legend

GET read	POST create	PUT update	DELETE remove
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1. Authentication

Everything starts here. Register an account and log in to receive an access token.

Method	Endpoint	Role	Request Body	Expected Response
POST	/api/auth/register	Public	FirstName, LastName, Email, Password, Phone, DateOfBirth	201 Created – User registered successfully
POST	/api/auth/login	Public	Email, Password	200 OK – Auth token and user details

2. User Profile

The authenticated user reads and updates their own details. Role is returned, never edited by the client.

Method	Endpoint	Role	Request Body	Expected Response
GET	/api/users/profile	Participant or Organiser	None	200 OK – User profile (including role)
PUT	/api/users/profile	Participant or Organiser	FirstName, LastName, Phone, DateOfBirth	200 OK – Updated profile

3. Events

The heart of RaceDay. Anyone can browse races; only the owning organiser may change them.

Method	Endpoint	Role	Request Body	Expected Response
GET	/api/events	Public	None	200 OK – List of events
GET	/api/events/{id}	Public	None	200 OK – Event details
POST	/api/events	Organiser	EventName, EventDate, Location, Description, EntryFee, Status	201 Created – Event created
PUT	/api/events/{id}	Organiser (owner)	EventName, EventDate, Location, Description, EntryFee, Status	200 OK – Event updated
DELETE	/api/events/{id}	Organiser (owner)	None	204 No Content – Event deleted

4. Categories

Categories describe the distances inside an event. They cannot be deleted once enrolments exist.

Method	Endpoint	Role	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Public	None	200 OK – List of categories
GET	/api/categories/{id}	Public	None	200 OK – Category details
POST	/api/events/{eventId}/categories	Organiser (event owner)	CategoryName, DistanceKm, Description	201 Created – Category created
PUT	/api/categories/{id}	Organiser (event owner)	CategoryName, DistanceKm, Description	200 OK – Category updated
DELETE	/api/categories/{id}	Organiser (event owner)	None	204 No Content – Category deleted (if no enrolments)

5. Event Enrolments

Participants secure race entries. Enrolments start as **pending** and confirm once payment is recorded.

Method	Endpoint	Role	Request Body	Expected Response
GET	/api/enrolments/my	Participant	None	200 OK – Enrolments with event and category details
GET	/api/enrolments/{id}	Participant or Organiser	None	200 OK – Enrolment details
POST	/api/enrolments	Participant	EventCategoryID	201 Created – Enrolment created with status 'pending'
PUT	/api/enrolments/{id}	Organiser (event owner) or Participant (own, cancel only)	EnrolmentStatus	200 OK – Enrolment updated
DELETE	/api/enrolments/{id}	Participant (owner) or Organiser	None	204 No Content – Enrolment cancelled
GET	/api/events/{eventId}/enrolments	Organiser (event owner)	None	200 OK – Enrolments with participant details

6. Results

Organisers capture finish times after the race. The leaderboard is public; personal results are private.

Method	Endpoint	Role	Request Body	Expected Response
GET	/api/results	Public	None	200 OK – Results with participant names
GET	/api/results/{id}	Public	None	200 OK – Result details
POST	/api/results	Organiser (event owner)	EnrolmentID, FinishTime, Position, ResultStatus	201 Created – Result recorded
PUT	/api/results/{id}	Organiser (event owner)	FinishTime, Position, ResultStatus	200 OK – Result updated
DELETE	/api/results/{id}	Organiser (event owner)	None	204 No Content – Result deleted
GET	/api/users/me/results	Participant	None	200 OK – Results for this participant

7. Payments

Each enrolment carries at most one payment record, linked one-to-one by EnrolmentID.

Method	Endpoint	Role	Request Body	Expected Response
POST	/api/payments	Participant	EnrolmentID, PaymentMethod	201 Created – Payment recorded with status 'paid'

Method	Endpoint	Role	Request Body	Expected Response
GET	/api/payments/{id}	Participant (owner) or Organiser	None	200 OK – Payment details
GET	/api/enrolments/{enrolmentId}/payment	Participant or Organiser	None	200 OK – Payment details

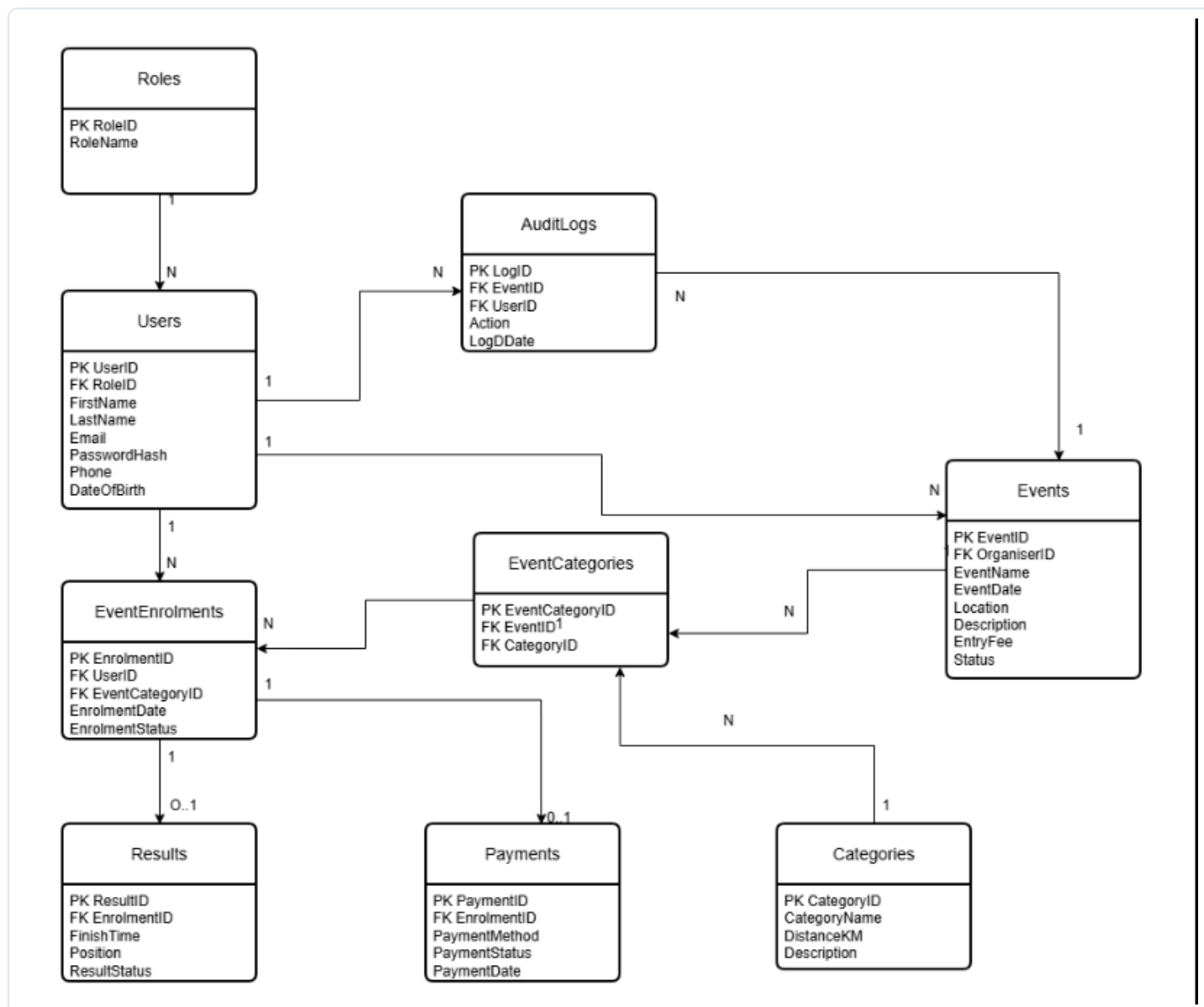
8. Audit Logs

Every meaningful change to an event is written to AuditLogs with the acting user, action and timestamp.

Method	Endpoint	Role	Request Body	Expected Response
GET	/api/events/{eventId}/logs	Organiser (event owner)	None	200 OK – Audit log entries for the event

9. Database Model

The API maps directly onto the RaceDay relational schema below.



RaceDay entity relationship diagram — Roles, Users, Events, EventCategories, Categories, EventEnrolments, Results, Payments and AuditLogs.

Table roles at a glance

- **Roles** → **Users** one-to-many. RoleID decides whether a user is a Participant or an Organiser.
- **Users** → **Events** one-to-many through OrganiserID. Only the organiser who created an event may modify it.
- **Events** ↔ **Categories** many-to-many resolved by **EventCategories**; a participant always enrolls against an EventCategoryID.
- **EventEnrolments** → **Results** and → **Payments** are both zero-or-one relationships.
- **AuditLogs** references both the Event and the User responsible for the action.

Design Notes for Student Developers

1. Role enforcement

Middleware inspects the token claim (**role: organiser | participant**). Endpoints such as **/api/users/profile** and **/api/enrolments/my** always resolve to the authenticated user — never trust a client-supplied ID for ownership.

2. Ownership checks

- Participants may view or cancel only their own enrolments, payments and results.
- Organisers have full control of the events they created, and nothing else.
- Enforce ownership in the query: **WHERE UserID = :authUserId** or **WHERE OrganiserID = :authUserId**.

3. Enrolment workflow

- Participant selects an event, then an EventCategoryID within it.
- The system confirms the event Status is open before creating the row.
- The enrolment is created as **pending**, then moves to **confirmed** after payment, or **cancelled**.
- Prevent duplicate enrolments for the same user and event category.

4. Results workflow

- Results may only be captured against a confirmed enrolment after the event date.
- Validate FinishTime as **HH:MM:SS** and keep Position unique per category.
- ResultStatus supports **finished**, **DNF** (did not finish) and **DNS** (did not start).

5. Validation and cascade rules

- Use ISO dates: **YYYY-MM-DDTHH:MM:SS**. EntryFee and DistanceKm must be positive.
- A category may not be deleted while enrolments reference it.
- Deleting an event cascades to its event categories, enrolments, payments and results — prefer a soft delete via Status.

START SMALL — build Authentication, Events and Categories first. Once that vertical slice works end to end, layer on Enrolments, Payments, Results and finally Audit Logs.